1.

Simulate a DNS resolution failure on your local machine by editing the hosts file to point www.spotify.com to a wrong IP address, then try to ping the domain and document the error message you receive.

Procedure:

Step 1: Edit the Hosts File

1. I opened Notepad as Administrator.
2. I opened the hosts file from:
   C:\Windows\System32\drivers\etc\hosts
3. I added the following entry:

192.0.2.1 www.spotify.com

---

4. I saved the file.

Step 2: Test Using Ping

1. I opened Command Prompt (CMD).
2. I entered the following command:

---

ping www.spotify.com

---

3. I observed that the domain resolved to the incorrect IP address 192.0.2.1, causing the ping to fail.

Expected Error Message:

Pinging www.spotify.com [192.0.2.1]

Request timed out.

Step 3: Restore the Original Settings

I removed the added entry from the hosts file and ran:

---

ipconfig /flushdns

Result: Successfully simulated a DNS resolution problem by mapping www.spotify.com to an incorrect IP address and testing it using the ping command.

2.

Configure a DHCP server in a network simulator (such as Cisco Packet Tracer or GNS3) and intentionally set an incorrect default gateway; connect a client device and verify the assigned gateway, then describe how you would identify and fix this issue.

Procedure:

Step 1: Configure DHCP Server

5. I opened Cisco Packet Tracer and created a network with a router, switch, server, and PC.
6. I configured the DHCP server with the following settings:
    - IP Address: 192.168.1.10
    - Subnet Mask: 255.255.255.0
    - Default Gateway: 192.168.2.1 (Incorrect)
    - DHCP Pool: 192.168.1.100 onwards.

Step 2: Connect the Client

1. I connected the PC to the network.
2. I selected Desktop → IP Configuration → DHCP.
3. The PC received an IP address and the incorrect gateway 192.168.2.1.

Step 3: Identify the Problem

I opened Command Prompt and used:

---

ipconfig /all

I observed that the default gateway was incorrect because it belonged to a different subnet.

Step 4: Fix the Problem

1. I changed the DHCP default gateway to 192.168.1.1.
2. I renewed the client's IP configuration using:

---

ipconfig /release

ipconfig /renew

---

3. I checked the new gateway using ipconfig /all.

Result: Successfully simulated an incorrect DHCP default gateway, identified the issue, and corrected it by configuring the proper gateway address.

3.

Given a scenario where devices on your simulated network are unable to access websites by name but can by IP address, use the nslookup or dig tool to diagnose the DNS problem and write down the troubleshooting steps and solution.

procedure:

Step 1: Identify the Problem

7. I opened Command Prompt (CMD).
8. I tested internet connectivity using:

---

ping 8.8.8.8

---

9. The ping was successful, confirming IP connectivity was working.

Step 2: Diagnose DNS Using nslookup

1. I entered the following command:

---

nslookup www.google.com

---

2. I observed that the DNS lookup failed, indicating a possible DNS server problem.

Step 3: Check DNS Configuration

1. I used the following command:

---

ipconfig /all

---

2. I checked whether the correct DNS server address was configured.

Step 4: Fix the Problem

1. I configured a valid DNS server address, such as Google's DNS (8.8.8.8).
2. I cleared the DNS cache using:

---

ipconfig /flushdns

---

3. I tested the domain again using nslookup.

Result: Successfully identified the DNS resolution problem and fixed it by configuring a valid DNS server. Websites could now be accessed using their domain names.

4.

A device in your simulated network is not receiving an IP address from the DHCP server and shows an APIPA address (169.x.x.x). List the possible causes for this and document the step-by-step process you would use to resolve it. Hint: Check for network connectivity, DHCP scope exhaustion, and server availability.

Procedure:

Step 1: Check Network Connectivity

1. I checked whether the network cable was properly connected.
2. I verified that the client was connected to the correct switch or network.

Step 2: Check DHCP Server

1. I checked whether the DHCP server was running.
2. I verified the DHCP pool and available IP addresses.
3. I checked the default gateway and subnet mask settings.

Step 3: Renew IP Address

I opened Command Prompt and entered:

---

ipconfig /release

ipconfig /renew

This releases the old IP address and requests a new one from the DHCP server.

Step 4: Verify the IP Address

I used:

---

ipconfig /all

I checked whether the device received a valid IP address instead of 169.254.x.x.

Solution:

I restored network connectivity, checked DHCP server availability, and ensured that sufficient IP addresses were available in the DHCP pool**.**

Result: Successfully identified the possible causes of DHCP failure and followed troubleshooting steps to obtain a valid IP address.

5.

Create a table documenting at least three common DNS issues and three common DHCP issues you encountered or researched, including symptoms, possible causes, and recommended solutions for each.

Common DNS Issues

DNS Issue	Symptoms	Possible Causes	Solutions
DNS Resolution Failure	Websites don't open by name.	DNS server unavailable.	Configure a valid DNS server.
Incorrect IP Address	Website opens incorrectly.	Wrong DNS A record.	Correct the DNS record.
DNS Cache Problem	Old or incorrect website address appears.	Outdated DNS cache.	Run ipconfig /flushdns.

Common DHCP Issues

DHCP Issue	Symptoms	Possible Causes	Solutions
APIPA Address	Device receives 169.254.x.x.	DHCP server unavailable.	Check and restart DHCP service.
IP Pool Exhaustion	New devices cannot get IP addresses.	All available IPs are assigned.	Increase the DHCP address pool.
Incorrect Gateway	Device cannot access other networks.	Wrong gateway in DHCP settings.	Configure the correct default gateway.